MonkMode

AI-Powered Personal Growth Platform

FULL STACK	SECURE	DATA-DRIVEN
React • Express • MongoDB	Clerk auth • user isolation • Arcjet rate limits	Analytics • weekly reports • AI guidance

Project Summary

MonkMode is a full-stack web application that brings daily productivity, wellness, and reflection into one focused workspace. Users track journals, tasks, habits, goals, workouts, and nutrition; explore cross-module analysis; and chat with Ming for personalized AI guidance and weekly feedback.

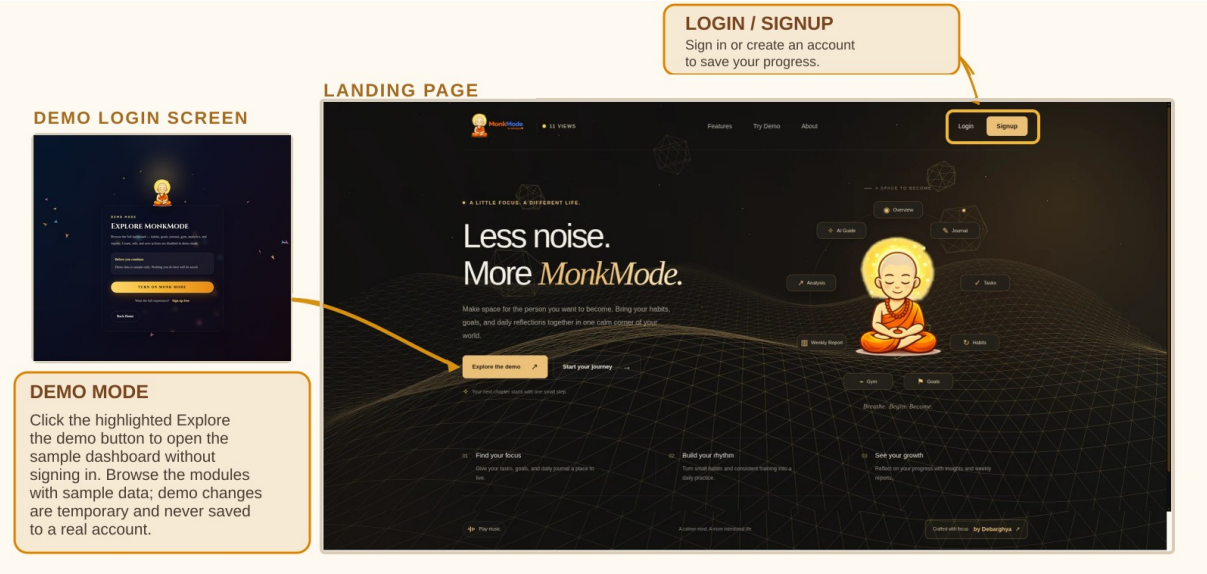
Motivation

I started MonkMode because, as a student, I struggled to stay consistent and focus on one task for long. I wanted to build a practical tool that helps people practice discipline each day and see their progress over time.

PURPOSE	PRODUCT APPROACH
Help students and self-improvement focused users build focus, self-control, and consistent daily habits.	Bring reflection, planning, habit and goal tracking, fitness, analytics, and Ming AI guidance together in one dashboard, with a read-only demo for exploration.

What Makes MonkMode Different

MonkMode brings journaling, tasks, habits, goals, fitness, and reflection into one workspace, then connects that activity to cross-module analytics, weekly feedback, and Ming AI guidance. Its read-only demo lets people explore the product without an account, while sample changes are never saved.



CORE FEATURES

Daily productivity —Journals, tasks, habits, and goals
Wellness —Workouts, nutrition, measurements, and progress photos
Analytics & AI —Cross-module insights, weekly reports, and Ming AI
Security & automation —Clerk auth, read-only demo, weekly reports, and Arcjet API protection. Smart Reminders with Nodemailer/node-cron are planned.

TECH STACK

Frontend —React + Vite
Backend —Node.js + Express
Database —MongoDB + Mongoose
Services —Clerk, Groq, Arcjet; Nodemailer and node-cron (planned)

Architecture & Application Flow

MonkMode follows a **three-tier architecture**. The React client attaches a Clerk session token to requests. Express verifies the request, applies business rules and rate limits, then reads or writes records owned by the authenticated user.

CLIENT	→	APPLICATION API	→	DATA & AI
React + Vite React Router • Axios Clerk session		Express controllers Validation • rate limits Business logic		MongoDB / Mongoose Groq AI • Arcjet

AUTHENTICATION & DATA ISOLATION	SAFE DEMO MODE
Protected routes use Clerk. The backend maps identity to a MongoDB user, and user-owned records are scoped by userId.	Bundled sample data enables product exploration while persistent API writes remain blocked.
DATA INTEGRITY	**RESILIENT REQUEST HANDLING**
Indexes and unique constraints support fast date-based views and prevent duplicate daily records.	Arcjet rate limits AI and write-heavy actions, with a local in-memory fallback.

Application lifecycle: sign in or demo → select a module → record activity → review trends → receive weekly guidance.

Application Screens

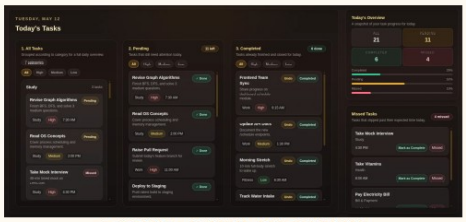
From daily actions to long-term growth: capture activity, review patterns, and use weekly guidance to decide what to improve next.



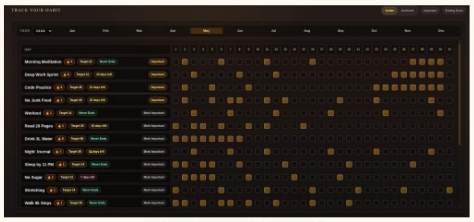
OVERVIEW DASHBOARD



JOURNAL



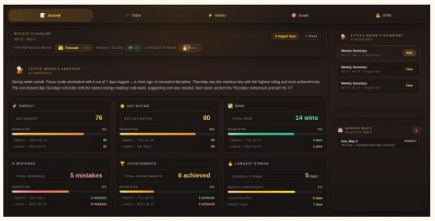
TASKS · DAILY OVERVIEW



HABIT TRACKING



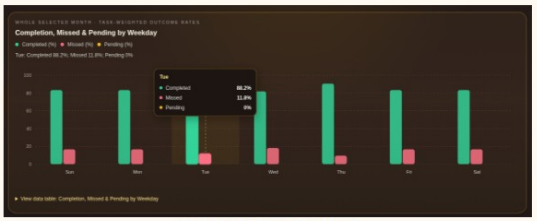
GOALS · PROGRESS



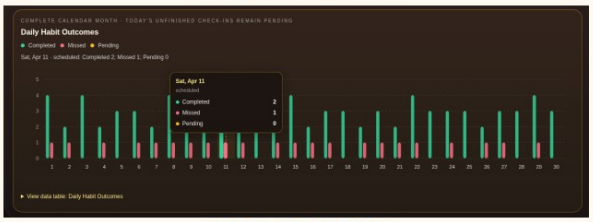
WEEKLY HABIT REPORT



MOOD DISTRIBUTION



TASK OUTCOMES



HABIT OUTCOMES



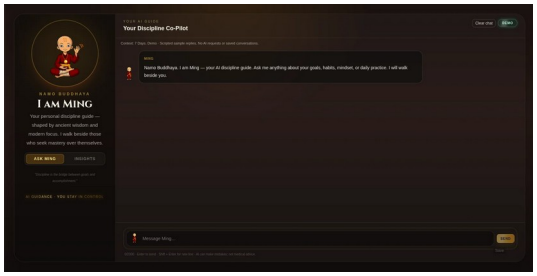
GYM PROGRESS

Core Capabilities

DAILY EXECUTION Journal, To-Do, and Habits transform daily actions into visible streaks, heatmaps, and consistency scores.	**GOALS & FITNESS** Goals, workouts, nutrition, measurements, and progress photos support long-term development.
ANALYTICS & REPORTS Analysis views reveal patterns across sleep, mood, energy, task outcomes, habits, and training volume.	**AI GUIDANCE** Ming provides contextual guidance from available user activity and supports cached weekly summaries.

Engineering Considerations

- **Security:** Clerk token verification, configurable CORS, schema validation, user-scoped data, and rate limits.
- **Performance:** lazy-loaded routes, indexed MongoDB queries, parallel data loading, and cached AI summaries.
- **Maintainability:** feature-oriented frontend folders and server separation across routes, controllers, models, middleware, and utilities.
- **Quality:** ESLint, backend regression tests, production builds, and explicit database environment separation.



AI Guru (Ming) — focused guidance based on real activity context.

Outcome: a cohesive, secure, and data-informed platform that helps users convert everyday effort into sustainable personal progress.

Live demo: https://monkmode.debarghya.org • Source: https://github.com/debarghya131/MonkMode